1.Download any popular Linux live boot ISO (such as Ubuntu or Kali Linux), create a bootable USB drive using Rufus or Balena Etcher, and verify that your system can boot from the USB into the live OS environment.

- Download Ubuntu ISO from https://ubuntu.com/download.
- Download Rufus from https://rufus.ie.
- Insert an 8 GB or larger USB drive.
- Open Rufus and select the USB drive.
- Select the Ubuntu ISO file.
- Click START to create the bootable USB.
- Restart the computer and press F12 to open the Boot Menu (on many Dell laptops).
- Select the USB drive.
- Click Try Ubuntu.

2.While running the live boot OS, use the file manager to access the Windows partition of your hard drive and locate the WhatsApp Desktop application's local data folder. List three types of user data you can view or copy from this folder.

- Boot the computer using Ubuntu USB.
- Open the Files application.
- Select the Windows partition.
- Open the Users folder.
- Select your Windows username.
- Open the AppData folder and find WhatsApp data.

Three Types of Data:

1. Media Files: Cached images and videos.
2. App Settings: WhatsApp application settings.
3. Cache Files: Temporary application data.

**3.Simulate an offline password attack: using the live boot OS, find the Windows SAM file (where user password hashes are stored), copy it to your USB drive, and explain in a text file how this file could be used in a password cracking tool like John the Ripper.

Hint: The SAM file is usually located in C:\Windows\System32\config.**

- Boot Ubuntu using a USB drive.
- Open the Windows partition.
- Identify the Windows SAM file location.
- For a safe classroom simulation, use a dummy hash file instead of copying real account credentials.
- Study how John the Ripper tests password guesses against supported hashes.

The SAM file stores Windows local account password hashes. Attackers may try to crack these hashes using password-guessing tools. Strong passwords and BitLocker encryption help protect against this attack.

**4.Boot into the live OS and use the built-in disk utility or terminal commands to mount and unmount a Windows NTFS partition. Write the exact commands or steps you used in a text file.

Hint: For Ubuntu, you can use the 'mount' and 'umount' commands in the terminal.**

```bash
lsblk -f
```

Step 2: Create a folder.

```bash
sudo mkdir -p /mnt/windows
```

Step 3: Mount the Windows partition in read-only mode.

```bash
sudo mount -t ntfs3 -o ro /dev/sda3 /mnt/windows
```

Step 4: View files.

```bash
ls /mnt/windows
```

Step 5: Unmount the partition.

```bash
sudo umount /mnt/windows
```

Expected Output:

```
Windows
Users
Program Files
```

5.Research using ChatGPT: Ask ChatGPT to list 3 real-world cyber attacks where attackers used a live boot OS for offline data access or password cracking. Summarize the most interesting case in your own words in a short paragraph.

Attack Name	Description
Cold Boot Attack (2008)	Researchers used reboot techniques to recover encryption keys from computer memory.
Evil Maid Attack (2009)	An attacker with physical access could modify the boot process to steal a disk password later.
Stoned Bootkit (2009)	A bootkit demonstrated how malicious boot code could compromise system security.

These are documented security demonstrations involving boot-level attacks, not three verified criminal incidents using live Linux USBs.

Most Interesting Attack: Evil Maid Attack

The Evil Maid Attack is a physical security attack. An attacker gets access to a laptop and modifies its boot software. When the owner starts the laptop and enters the disk password, the malicious software may steal it. This attack shows why physical security and boot protection are important.